

PRODUCT & BI ANALYST — DATA CHALLENGE

Growth runs on first purchases

Customers, churn and a free-shipping policy — and three moves to act on it.

Growth runs on first purchases

Revenue, Aug 2026 \$115K +111% year over year	From first-time buyers 84% of revenue, last 12 months	Buy again within a year 1 in 6 median wait: 412 days	Free shipping over \$100 -26% gross profit, simulated
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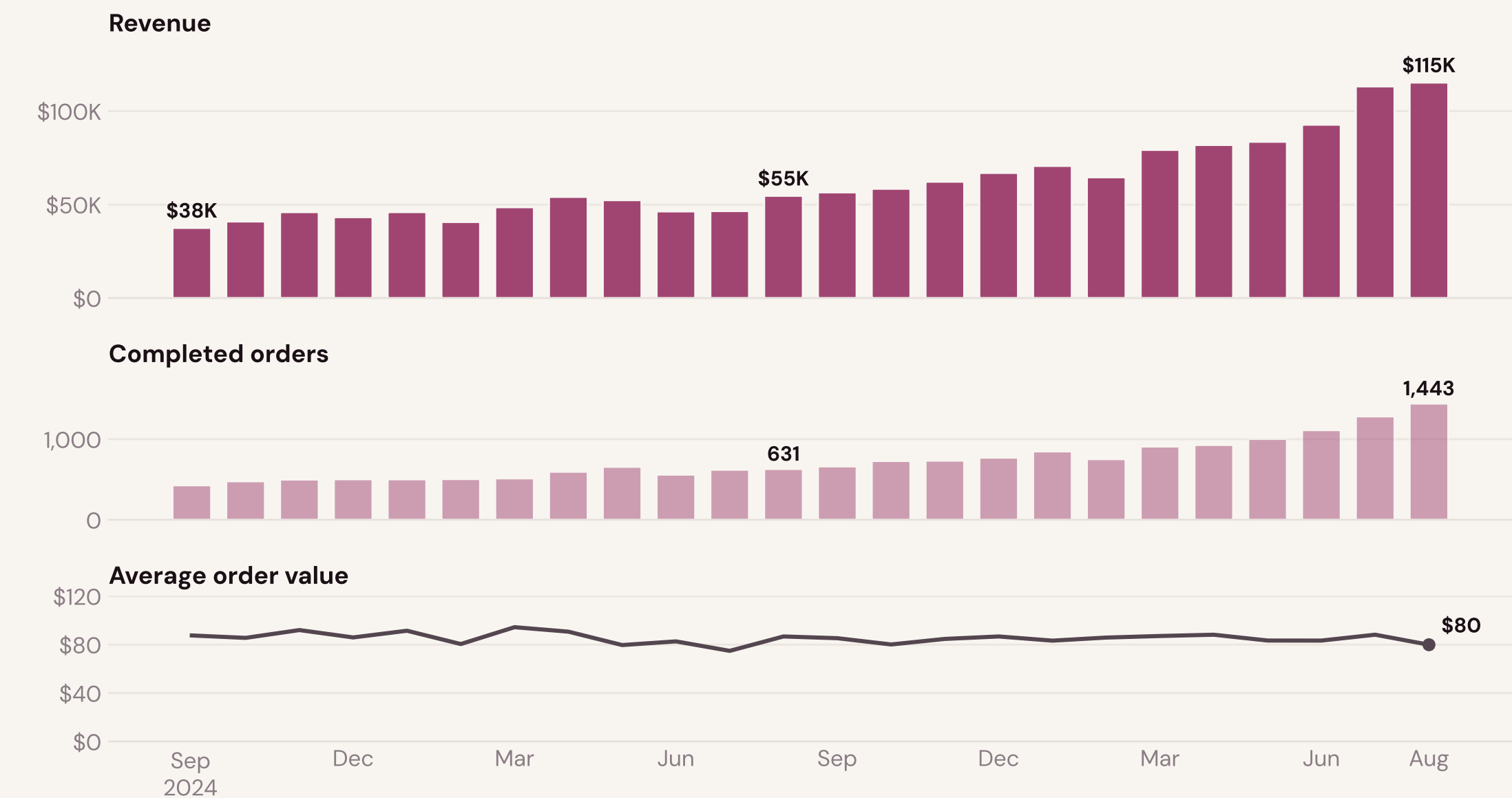
WHAT WE FOUND

- 1 Revenue **+111%** — all volume; basket size flat
- 2 **84%** of revenue from first-time buyers; few come back
- 3 Free shipping over \$100 **pays for baskets that already exist**

WHAT WE SHOULD DO

- 1 **Win the second purchase** in the first 60 days
- 2 **Free shipping on the second order** — US first
- 3 **Fix measurement** before scaling decisions

Revenue +111% — driven by volume, not basket

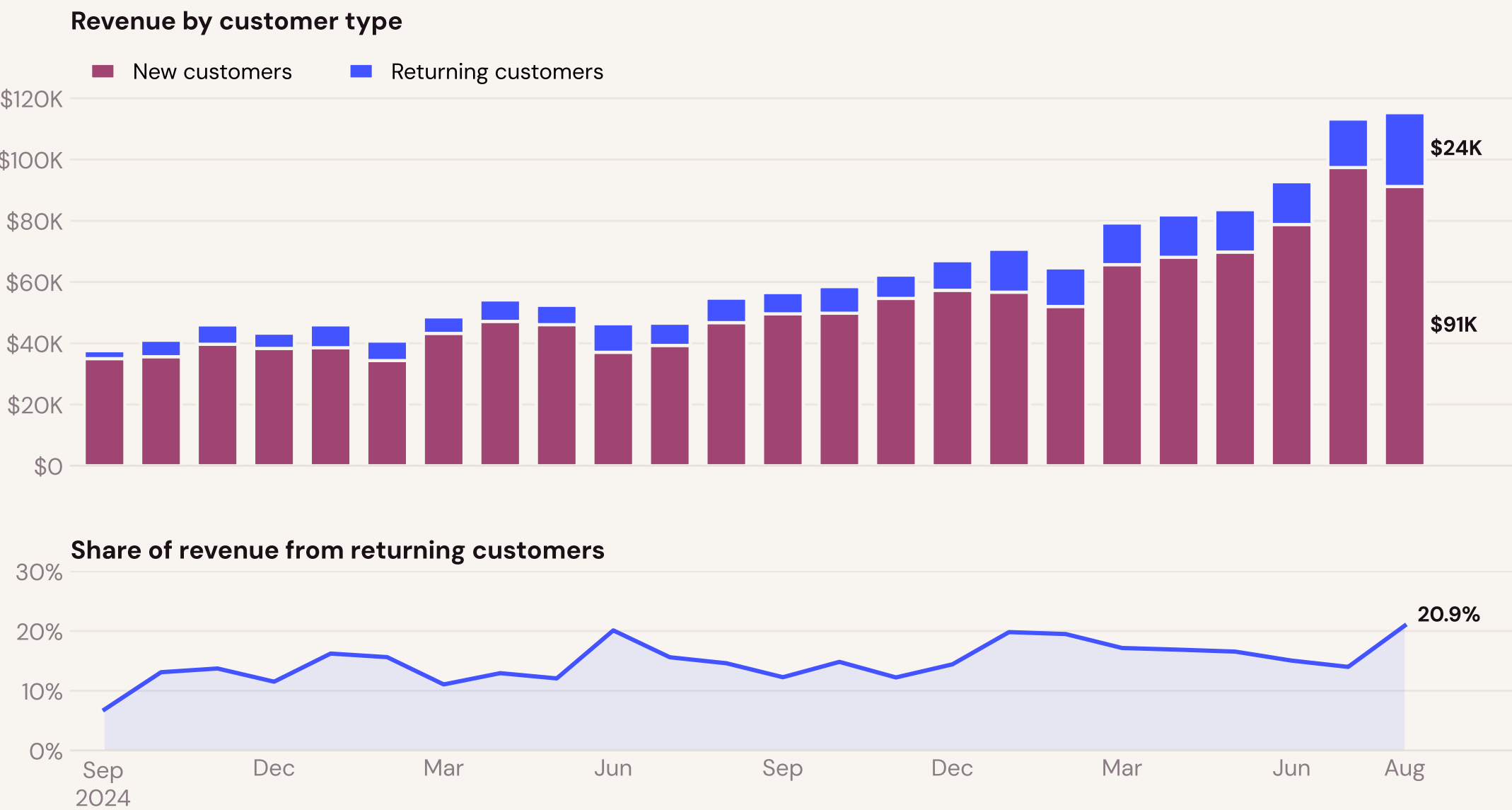


+129%
orders, year over year

\$75–\$94
average order value — flat for 2 years (28% of single orders top \$100)

\$946K
revenue, last 12 months (+70%)

84% of revenue comes from first-time buyers

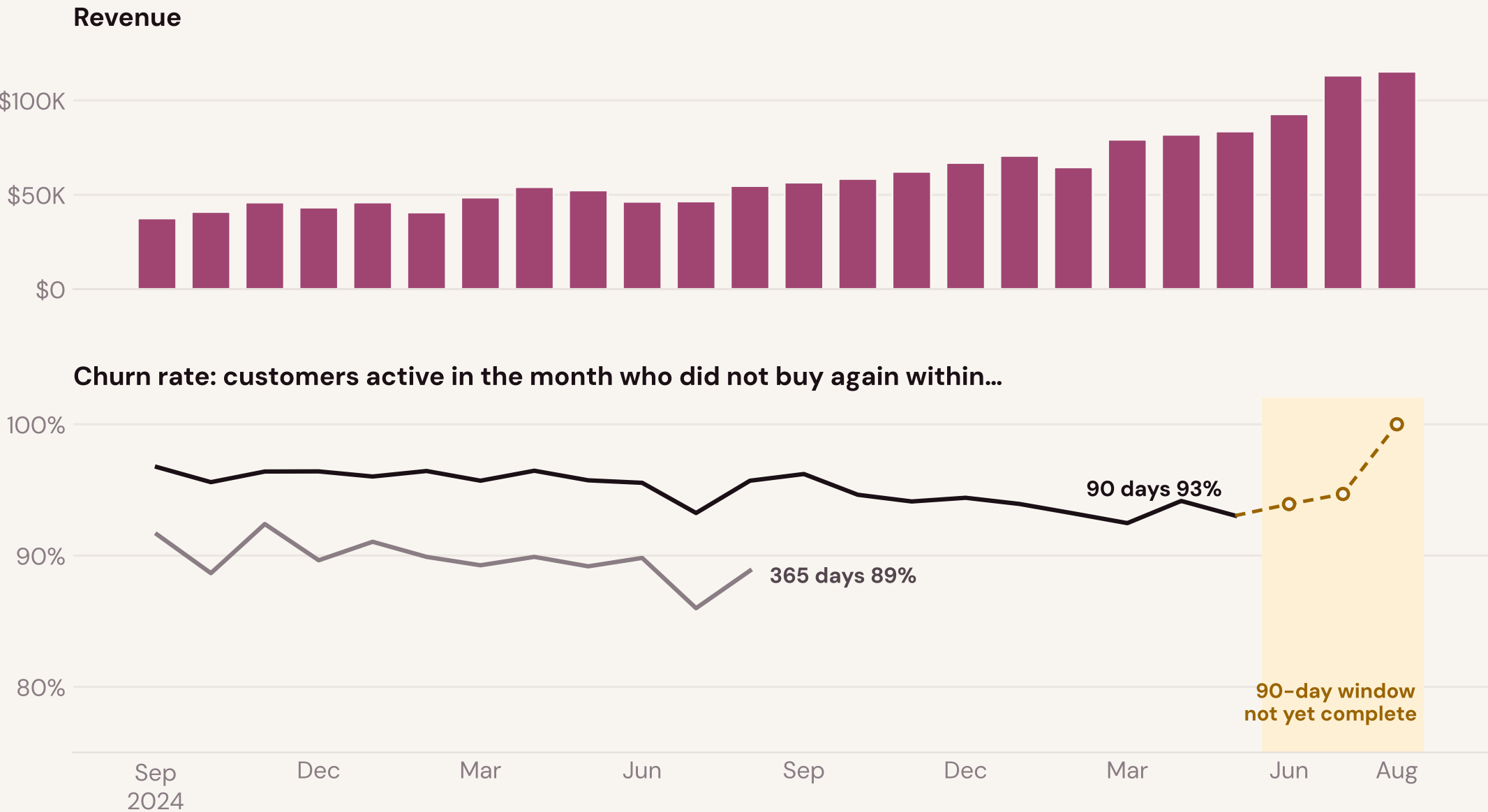


+12%
spend per returning customer vs new

-8%
revenue if new customers fall 10%

70%
of first-time buyers come from Search

Even over a full year, 90% don't come back



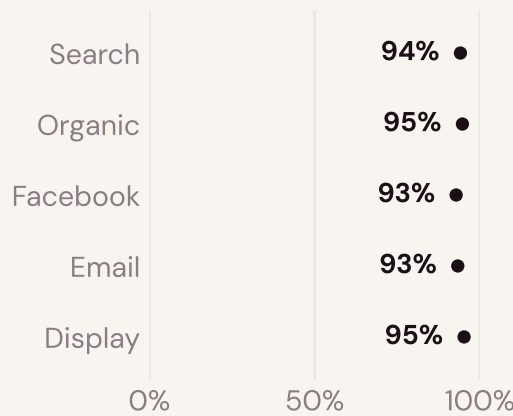
95.8% → 89.7%

churn at 90 vs 365 days

94.1% → 88.5%

if all valid orders count, not only 'Complete'

90-day churn by channel



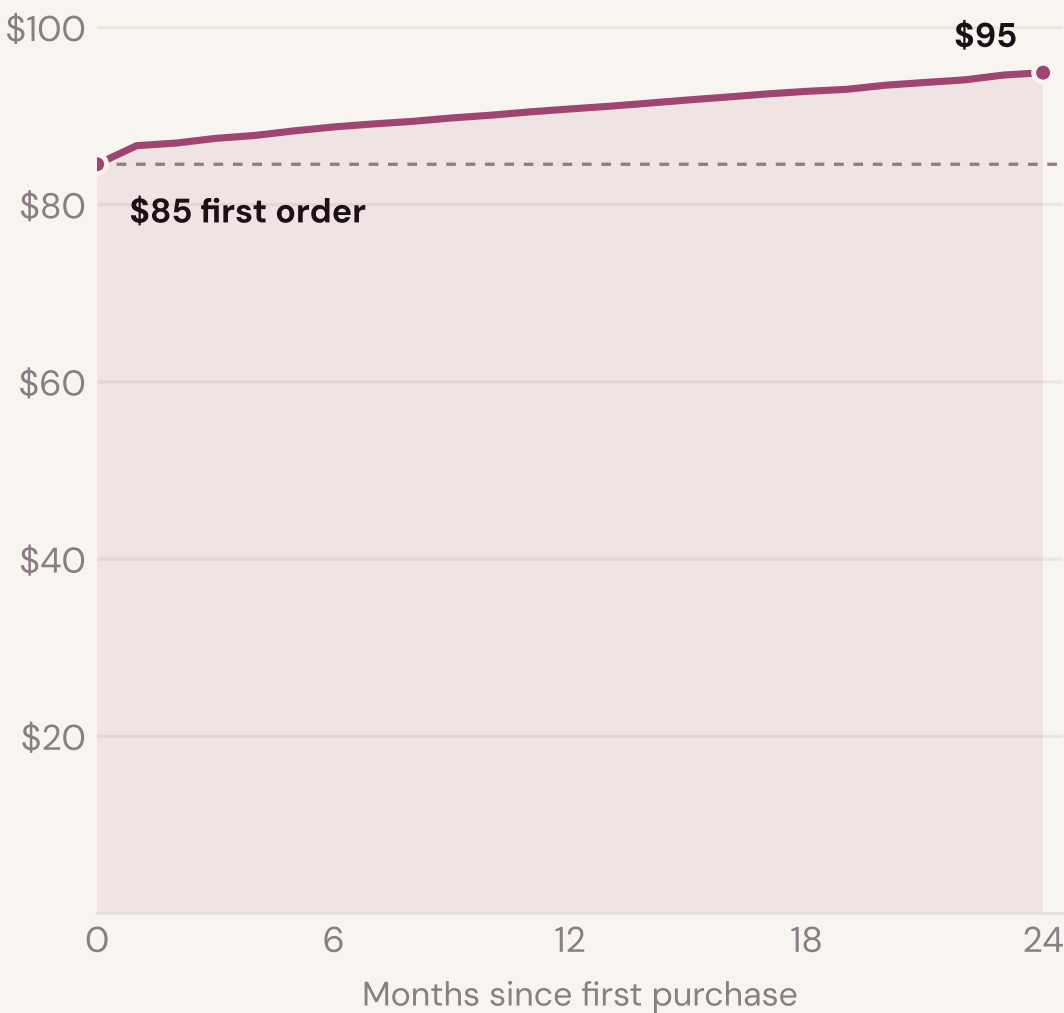
The first order is 89% of two-year value

Month 1 is the best chance of a second purchase.

Cohort retention — % buying again in month N

	M1	M2	M3	M4	M5	M6	M7	M8
Jan 2025 (430)	2.8	0.5	0.9	0.2	0.7	0.7	0.2	0.2
Feb 2025 (430)	1.4	0.9	1.2	0.9	0.2	1.2	0.5	1.2
Mar 2025 (446)	3.1	0.5	0.9	1.4	1.6	0.2	0.7	0.2
Apr 2025 (504)	2.2	1.0	0.6	0.4	1.2	0.8	1.0	0.8
May 2025 (572)	3.3	1.1	0.3	1.4	1.2	0.3	0.2	1.4
Jun 2025 (469)	3.0	0.8	0.8	0.2	0.2	0.8	0.4	0.6
Jul 2025 (521)	4.6	1.7	1.1	0.6	0.8	1.1	0.8	1.1
Aug 2025 (537)	2.6	1.1	0.9	0.6	0.7	1.1	0.7	0.7
Sep 2025 (566)	2.6	0.5	0.9	1.1	0.5	0.7	0.5	1.1
Oct 2025 (613)	2.9	1.0	1.6	1.6	2.0	1.1	1.3	1.1
Nov 2025 (633)	3.6	0.6	2.5	1.1	0.6	0.5	1.7	1.3
Dec 2025 (659)	3.2	1.4	1.1	1.5	1.2	1.8	0.9	2.1
Jan 2026 (686)	3.6	1.3	1.3	2.0	0.9	1.8	1.5	
Feb 2026 (609)	4.9	1.8	0.8	1.8	0.7	2.0		

Cumulative spend per customer



Three moves to build a repeat base

1

Win the 2nd purchase in 60 days

Month 1 is the repeat peak (3.1% vs 0.9%).

TARGET	All first-time buyers — by timing, not channel
KPI	12-month repeat vs holdout (today 16.2%)
GUARDRAIL	Incentive cost per extra order

+3 pts ≈ **\$25K**/yr

2

Free shipping on the 2nd order

Not over \$100: 91% of those parcels go to baskets that already exist.

TARGET	US first-time buyers (23% of revenue)
KPI	90-day repeat vs holdout (today 4.4%)
GUARDRAIL	Gross profit after shipping

Breaks even at **+1.4 pts**

3

Fix measurement first

‘Complete’ is a random 25% of orders; 90 days is too short a churn window.

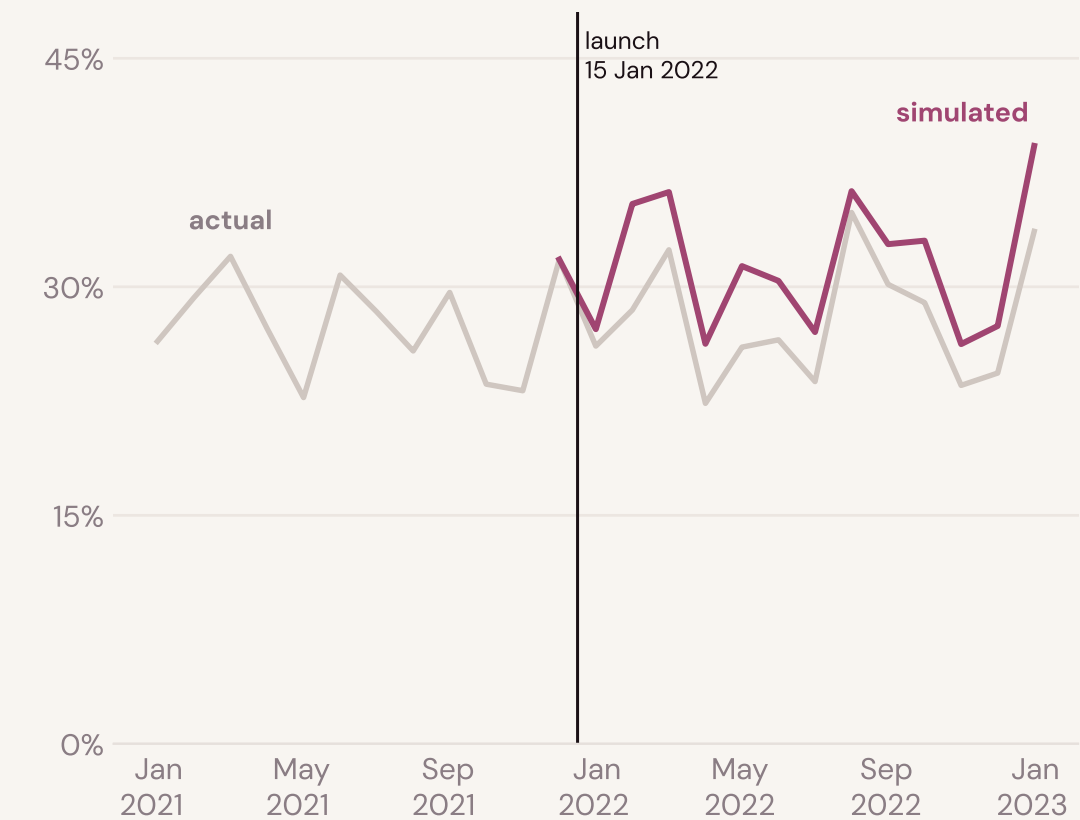
DO	Agree “sale” with finance · 12-month churn · automated QA
KPI	QA pass rate · data freshness

Makes 1 and 2 **measurable**

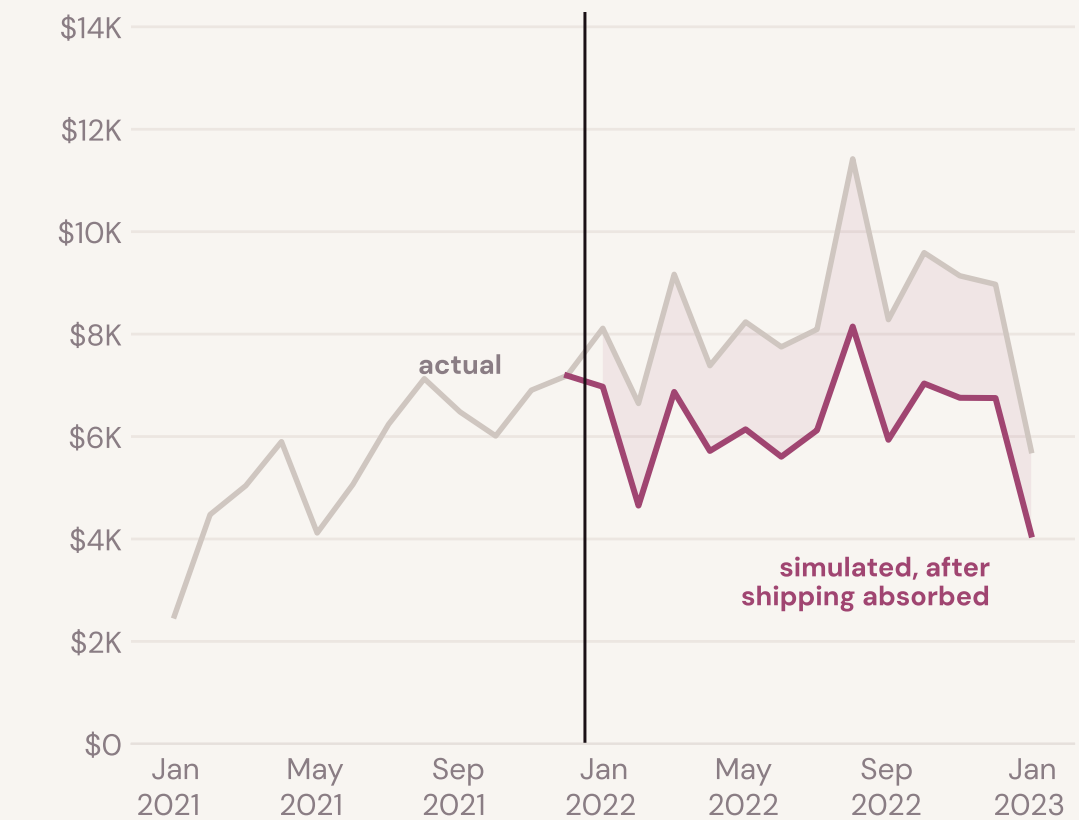
Free shipping over \$100 lifts revenue, cuts profit

Simulated impact: gross profit after shipping -26%.

Orders over \$100 — share of all orders



Gross profit per month — after shipping



28% → 31%

orders over \$100

+1.1%

revenue

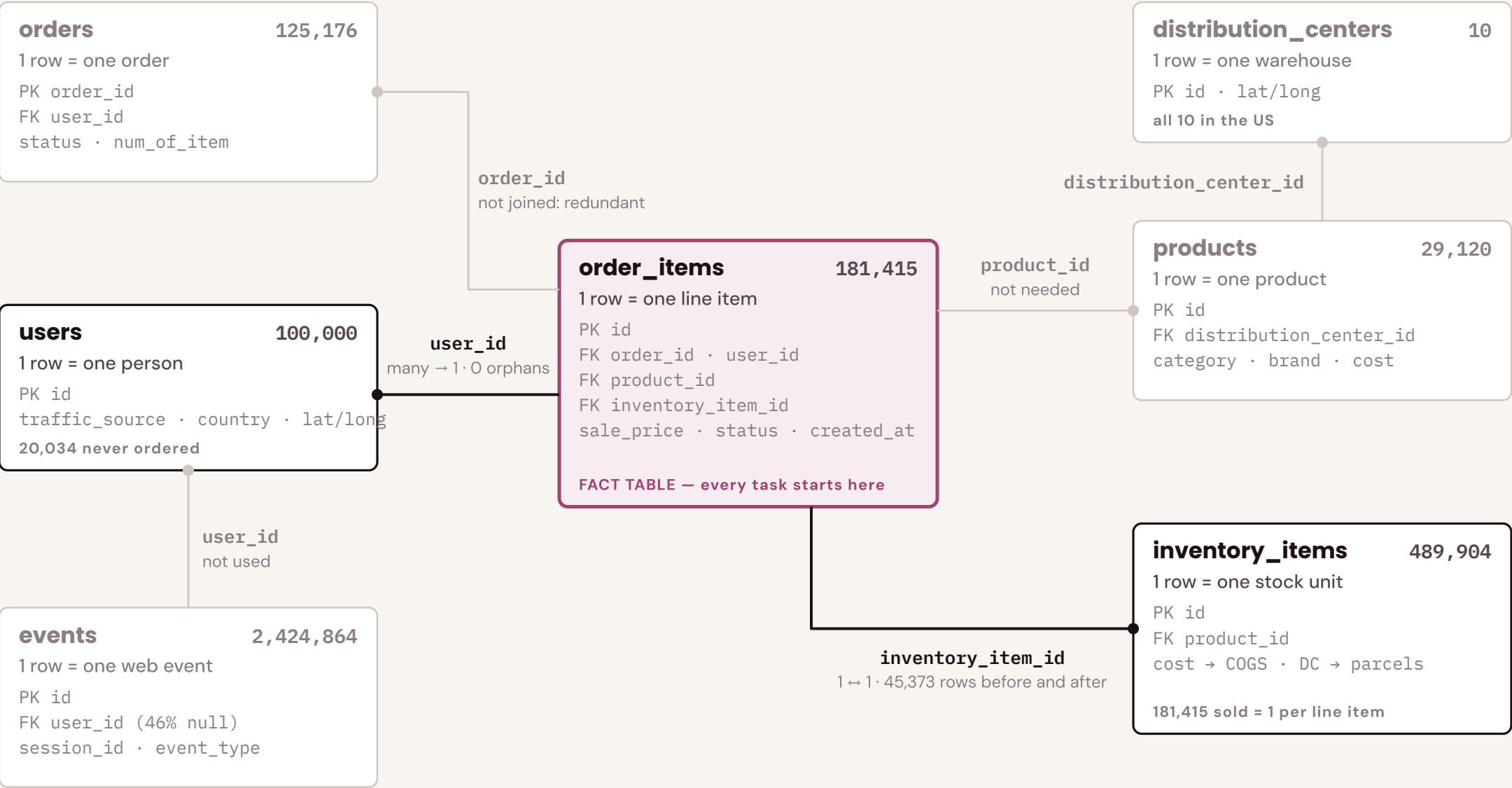
< \$1

break-even cost per parcel — 91% go to orders already over \$100

What each metric means — and when to change it

METRIC	DEFINITION USED	ALTERNATIVE	USE THE ALTERNATIVE WHEN...
Churn	Active in month M, no order in the next 90 days	12-month window	Purchases are infrequent (here: median 412 days)
Active customer	≥ 1 completed order in the month	Active in the last 12 months	The product is bought once or twice a year
New vs returning	New in the month of the first completed order	Returning = bought in last 12 months	The question is reactivation, not acquisition
Cohort	Month of first completed order	Signup month	Measuring activation, not repeat
Funnel	Signup → order → activated → 2nd order	Session: product → cart → purchase	Session data is real (here it is template-generated)
Product-change KPI	Primary: behaviour targeted · Guardrail: profit after its cost	Revenue alone	Never — revenue hid the free-shipping loss

One fact table, two joins, zero fan-out

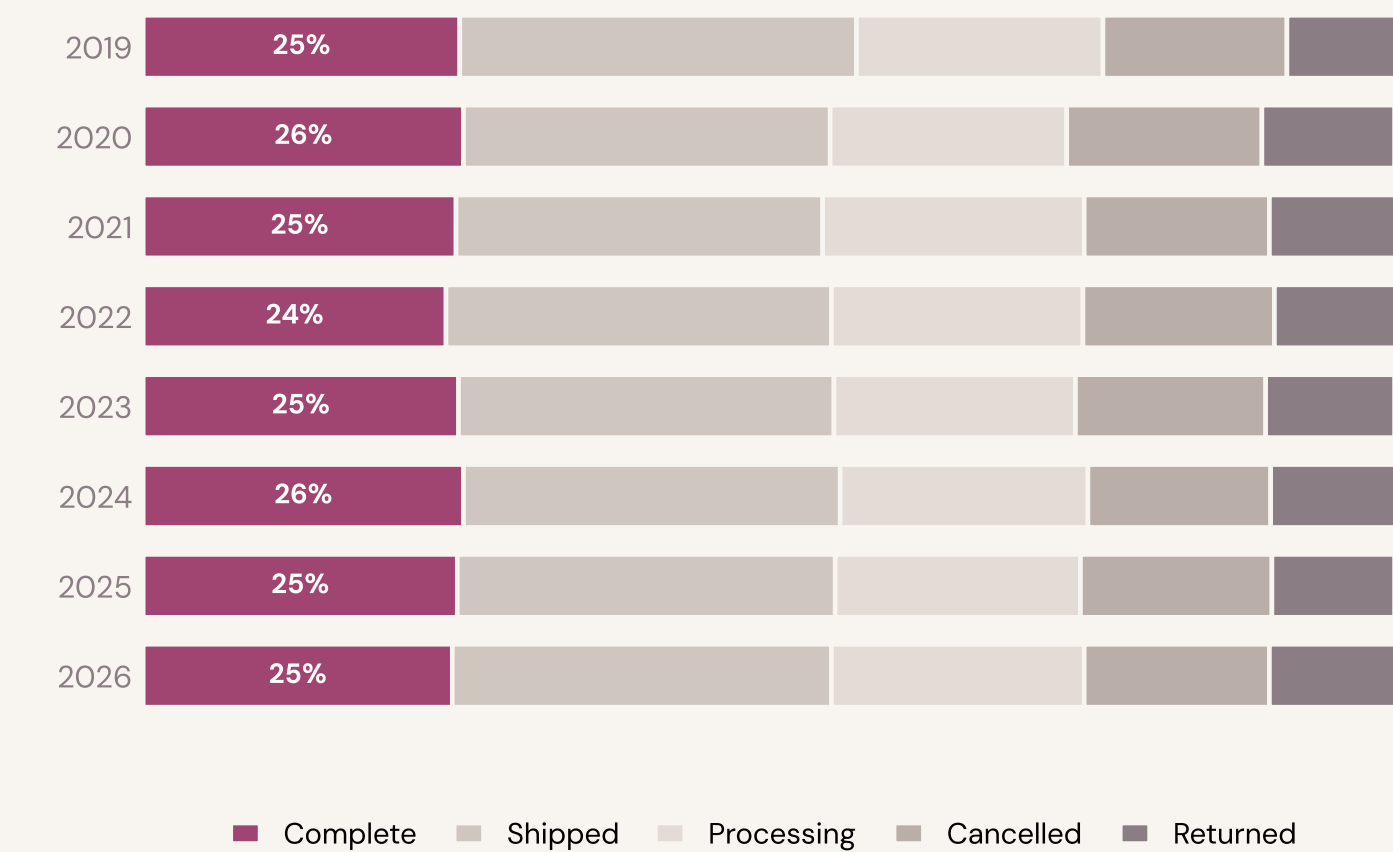


JOIN RULES

1. **Grain first** — one row per line item
2. **Count rows** before and after every join
3. **Skip joins** that add no columns (orders)
4. **Aggregate** before one-to-many joins

Four data traps, all handled

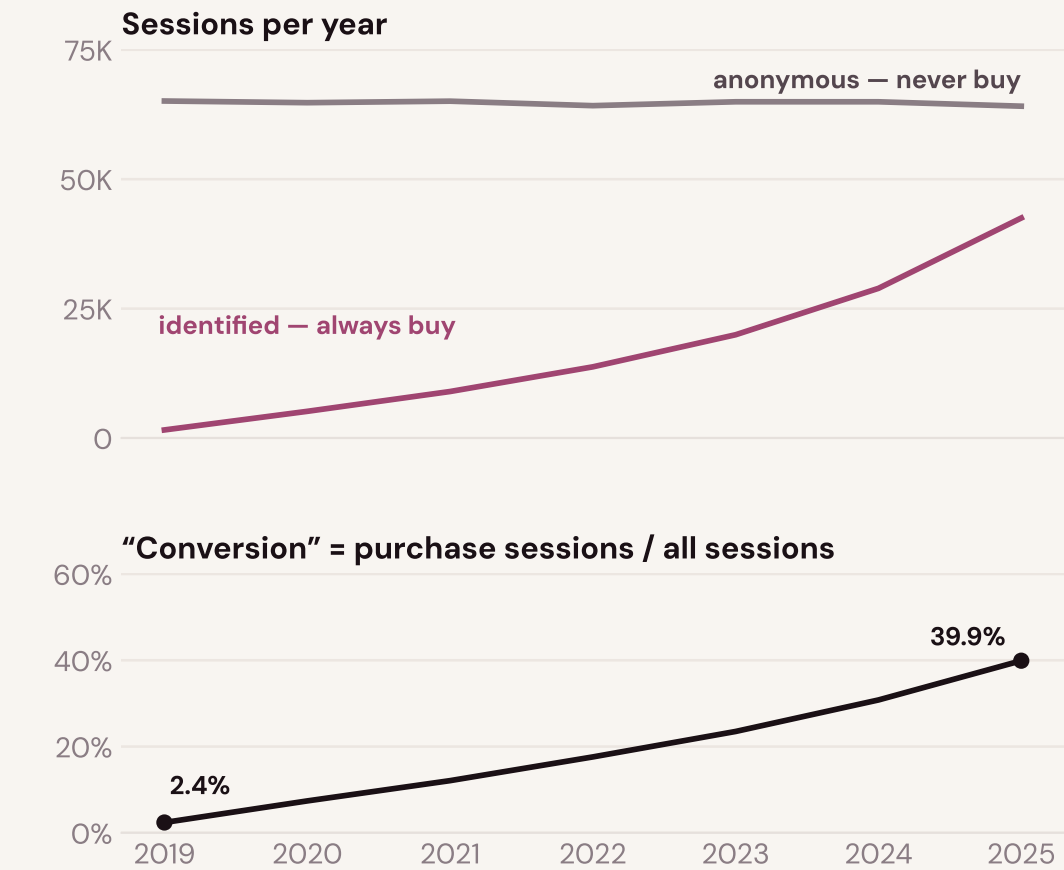
Order status by year — a 2019 order is still 'Shipped'



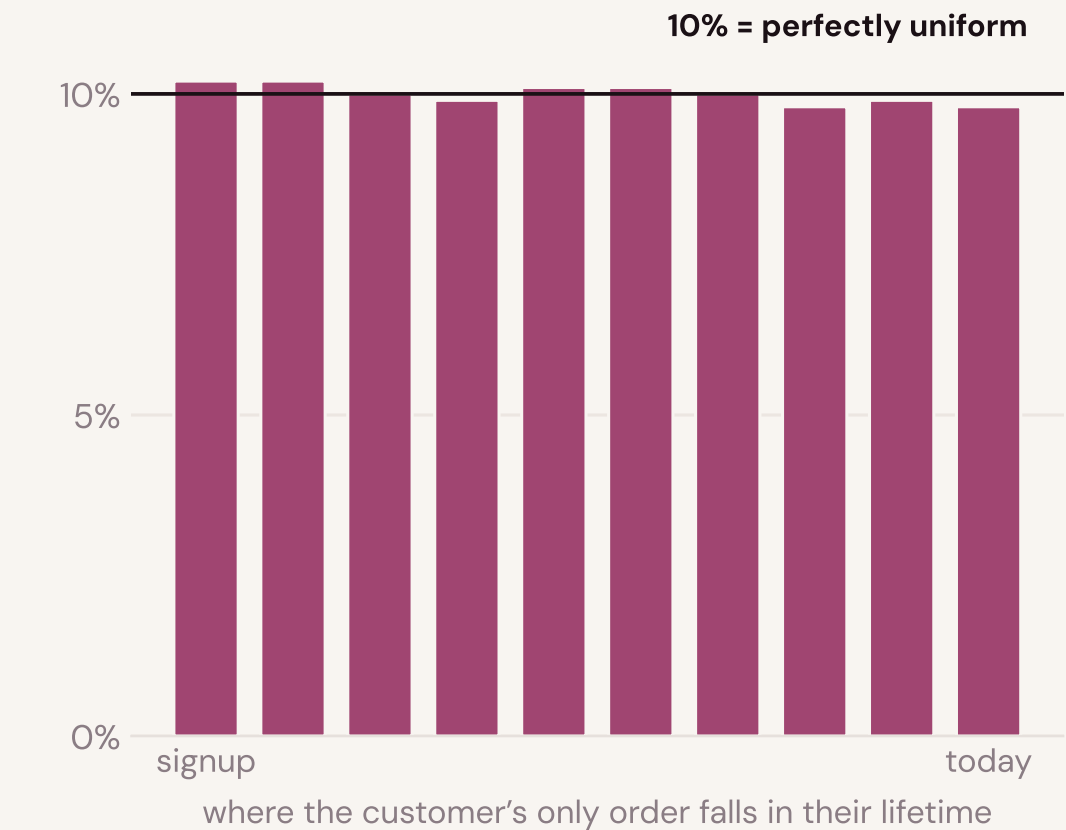
- 632 rows dated after today — excluded
- 25% of items 'Complete', every year — status is random
- 3 months with no full churn window — flagged, not reported
- 0 empty columns — every null is structural

Session data can't measure conversion

events — anonymous traffic is a constant



Order timing — orders fall at random between signup and today

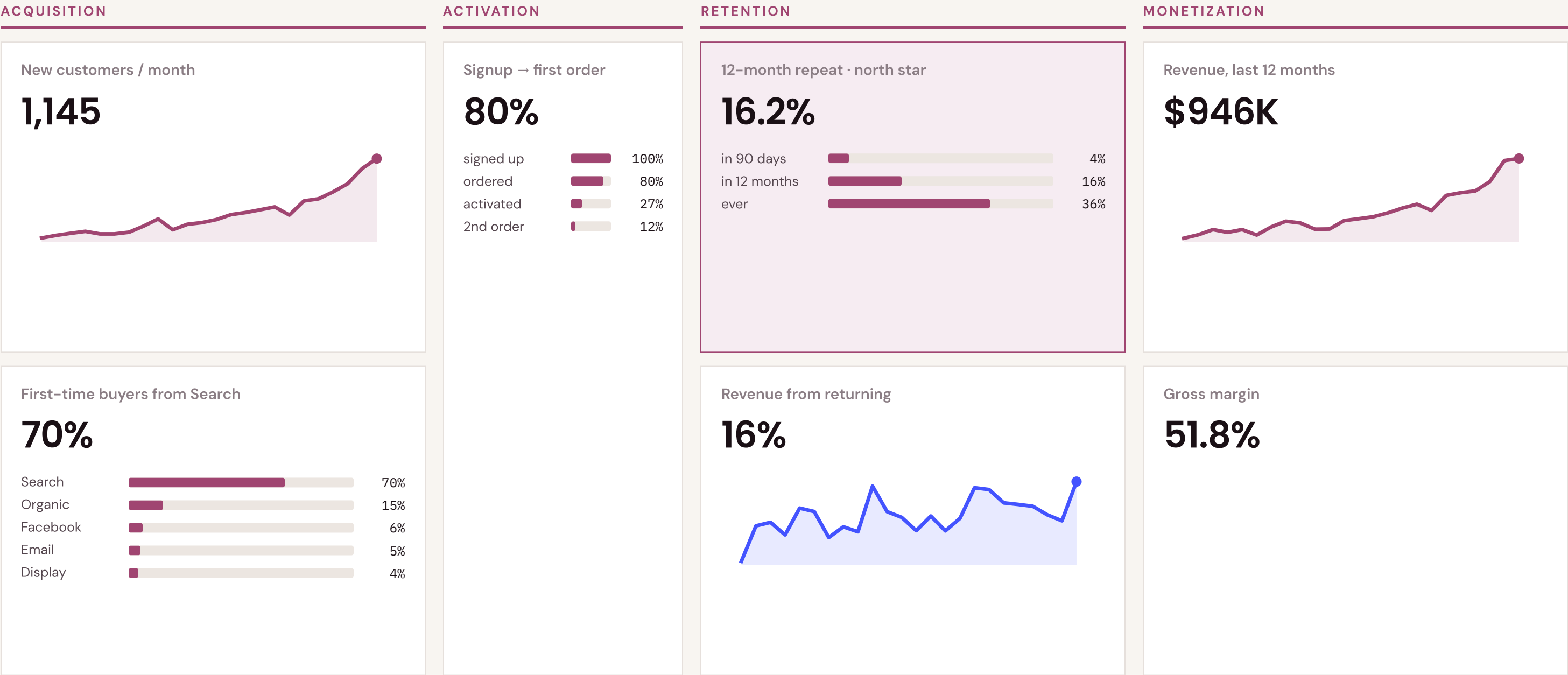


The funnel we use — per user

- 100 signed up
- 80 ordered
- 27 activated
- 12 2nd order

±0.6 pts
across channels

Seven metrics leadership should watch



DATA HEALTH	✓ last complete month Aug 2026	✓ 632 future rows excluded	✓ QA 4/4 passing	✓ as of 2026-09-24
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AI drafted the work; checks decided what stayed

USED FOR • Profiling tables, nulls and join keys • First drafts of every query • This deck, generated from the SQL outputs	CAUGHT BY CHECKS • 100% churn months = edge of the data • A metric that was 100% or 0% by construction • “Conversion” rising 17× on a constant denominator	HOW I VALIDATE • Row counts before / after joins • Identity checks that must return 0 rows • Same metric, two definitions
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EXAMPLE PROMPT

“Churn is ~100% for the last three months. Before treating it as real, what at the data boundary could cause it — and write the SQL check that tells an artifact from a real spike.”